

Roll: 220105

GitHub Profile: <https://github.com/mdraihan27>

GitHub Repo: <https://github.com/mdraihan27/svm-and-knn/tree/main>

Collab File Links:

1. SVM:
<https://colab.research.google.com/drive/1vGQ8JtyJgONfguFmxQuDLnO7B-QRXUy8?usp=sharing>
2. KNN:
<https://colab.research.google.com/drive/1B9RH5yANQ720LwrLDbmK7dYDtJNSR-0p?usp=sharing>

Dataset Links:

1. Diabetes Dataset (Used for SVM):
[Drive link](#)
[Kaggle link](#)
2. Heart Disease Dataset (Used for KNN):
[Drive link](#)
[Kaggle link](#)

Screenshots:

1. SVM

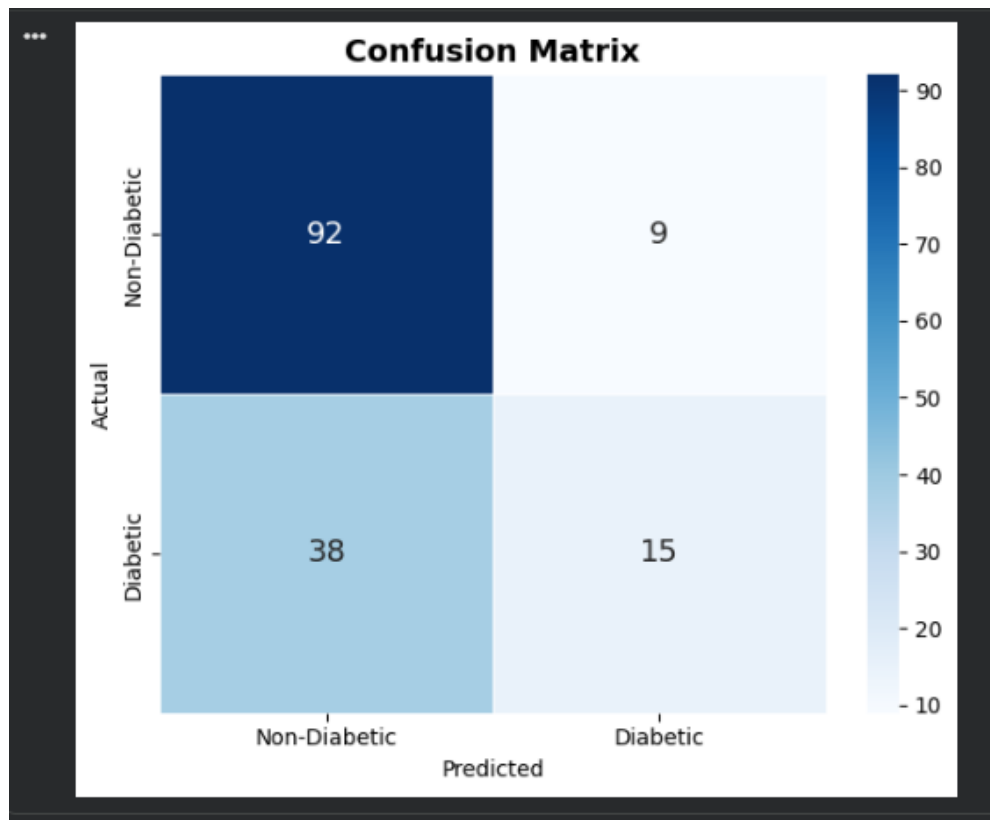


Figure 1: Confusion Matrix.

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                        EVALUATION METRICS
=====
Metric  Score
Accuracy 69.48%
Precision 62.5%
Recall 28.3%
F1-Score 38.96%
AUC 74.13%
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Full Classification Report:
              precision    recall  f1-score   support

Non-Diabetic    0.71      0.91      0.80       101
Diabetic        0.62      0.28      0.39        53

   accuracy          0.67
  macro avg          0.67
 weighted avg          0.68
  
```

Figure 2: Evaluation Metrics.

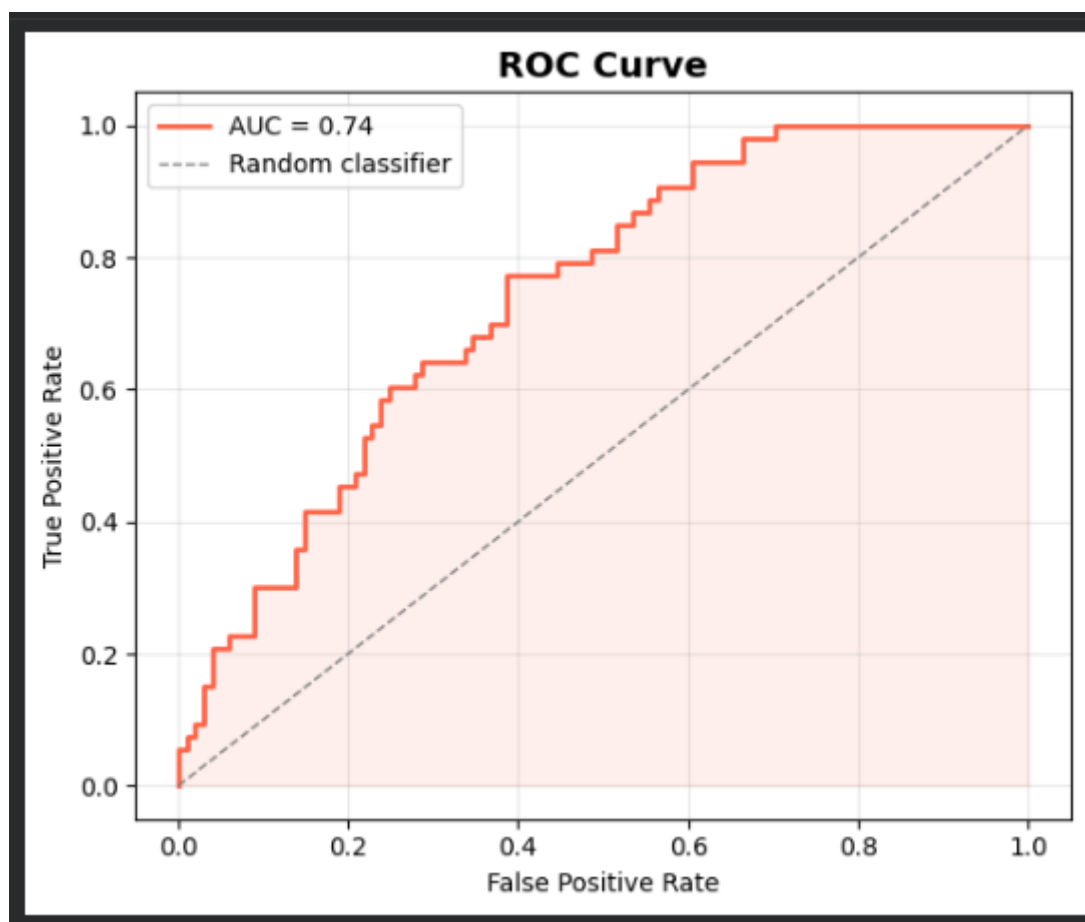


Figure 3: ROC curve.

SAMPLE PREDICTIONS (20 rows)						
	Actual	Predicted	Prob (Diabetic)	Prob (Non-Diabetic)	Correct	
0	Non-Diabetic	Non-Diabetic	48.11%	51.89%	YES	
1	Diabetic	Non-Diabetic	33.45%	66.55%	NO	
2	Diabetic	Non-Diabetic	31.46%	68.54%	NO	
3	Non-Diabetic	Non-Diabetic	22.13%	77.87%	YES	
4	Diabetic	Non-Diabetic	29.7%	70.3%	NO	
5	Non-Diabetic	Non-Diabetic	11.98%	88.02%	YES	
6	Diabetic	Diabetic	74.72%	25.28%	YES	
7	Diabetic	Non-Diabetic	23.0%	77.0%	NO	
8	Non-Diabetic	Non-Diabetic	29.23%	70.77%	YES	
9	Non-Diabetic	Non-Diabetic	24.5%	75.5%	YES	
10	Diabetic	Non-Diabetic	27.01%	72.99%	NO	
11	Non-Diabetic	Non-Diabetic	20.74%	79.26%	YES	
12	Diabetic	Non-Diabetic	25.56%	74.44%	NO	
13	Non-Diabetic	Non-Diabetic	35.12%	64.88%	YES	
14	Non-Diabetic	Non-Diabetic	25.6%	74.4%	YES	
15	Non-Diabetic	Non-Diabetic	8.15%	91.85%	YES	
16	Non-Diabetic	Non-Diabetic	22.06%	77.94%	YES	
17	Diabetic	Non-Diabetic	45.69%	54.31%	NO	
18	Diabetic	Diabetic	84.56%	15.44%	YES	
19	Diabetic	Non-Diabetic	19.72%	80.28%	NO	
Total Correct : 107 / 154						
Total Incorrect : 47 / 154						

Figure 4: Sample predictions.

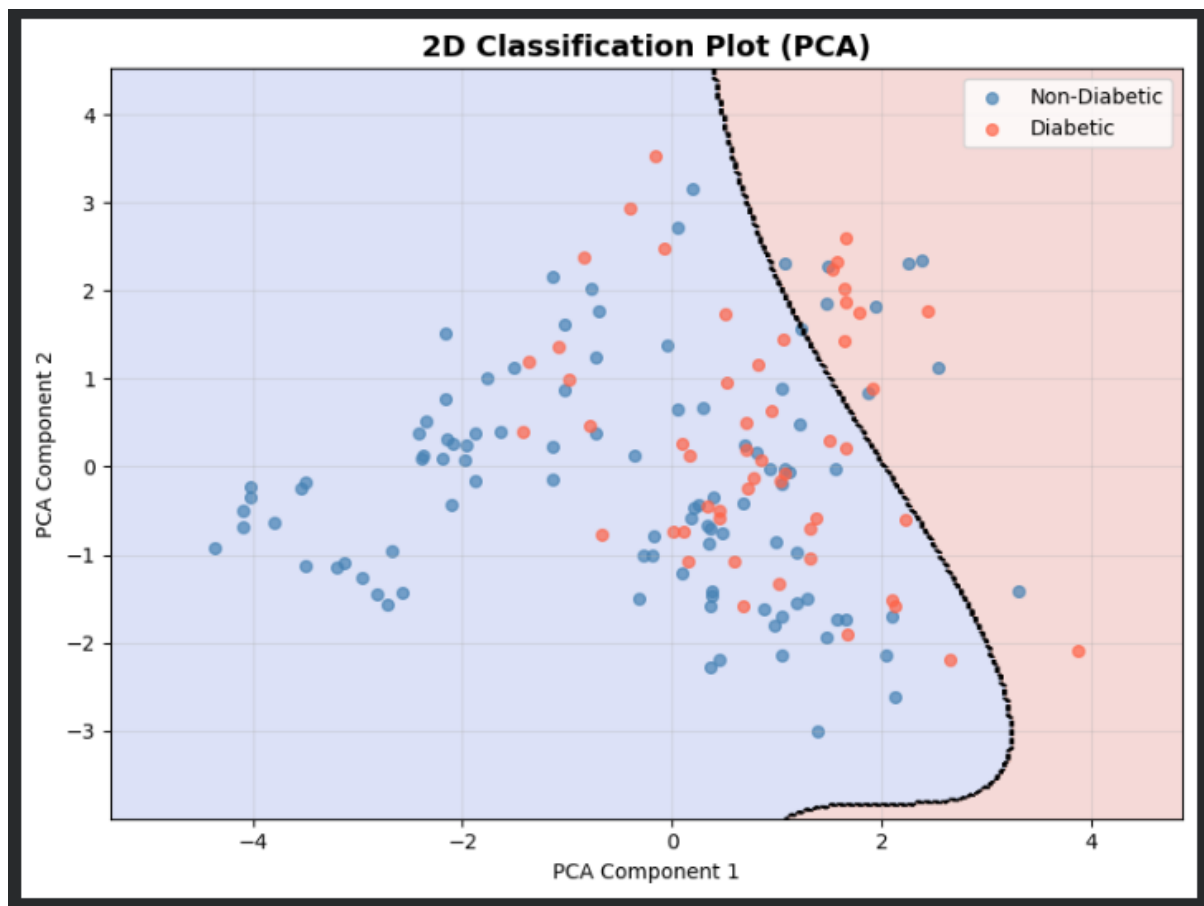


Figure 5: 2d classification plot.

2. KNN:

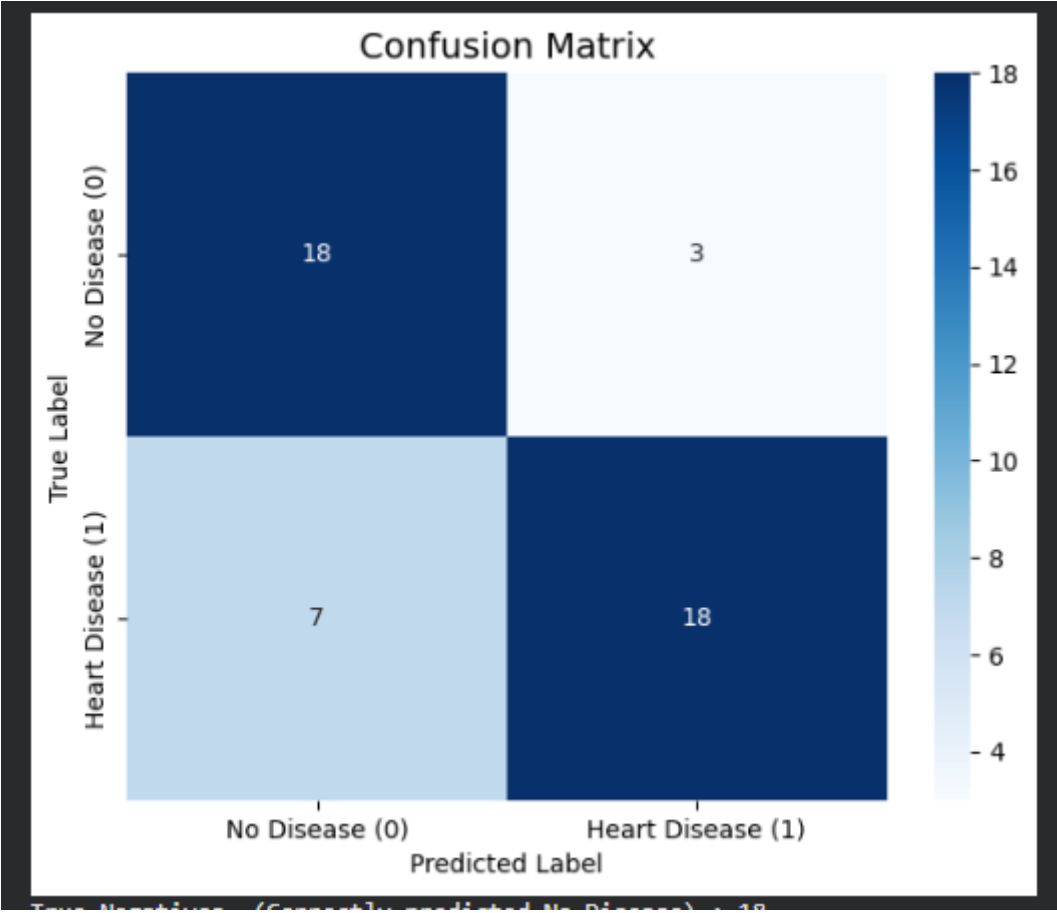


Figure 6: Confusion Matrix.

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...
Classification Report:
              precision    recall  f1-score   support

   No Disease (0)       0.72     0.86     0.78         21
   Heart Disease (1)    0.86     0.72     0.78         25

   accuracy              0.78         46
   macro avg              0.79         46
   weighted avg           0.78         46

Accuracy : 78.26%
AUC Score : 0.8495
```

Figure 7: Evaluation Metrics.

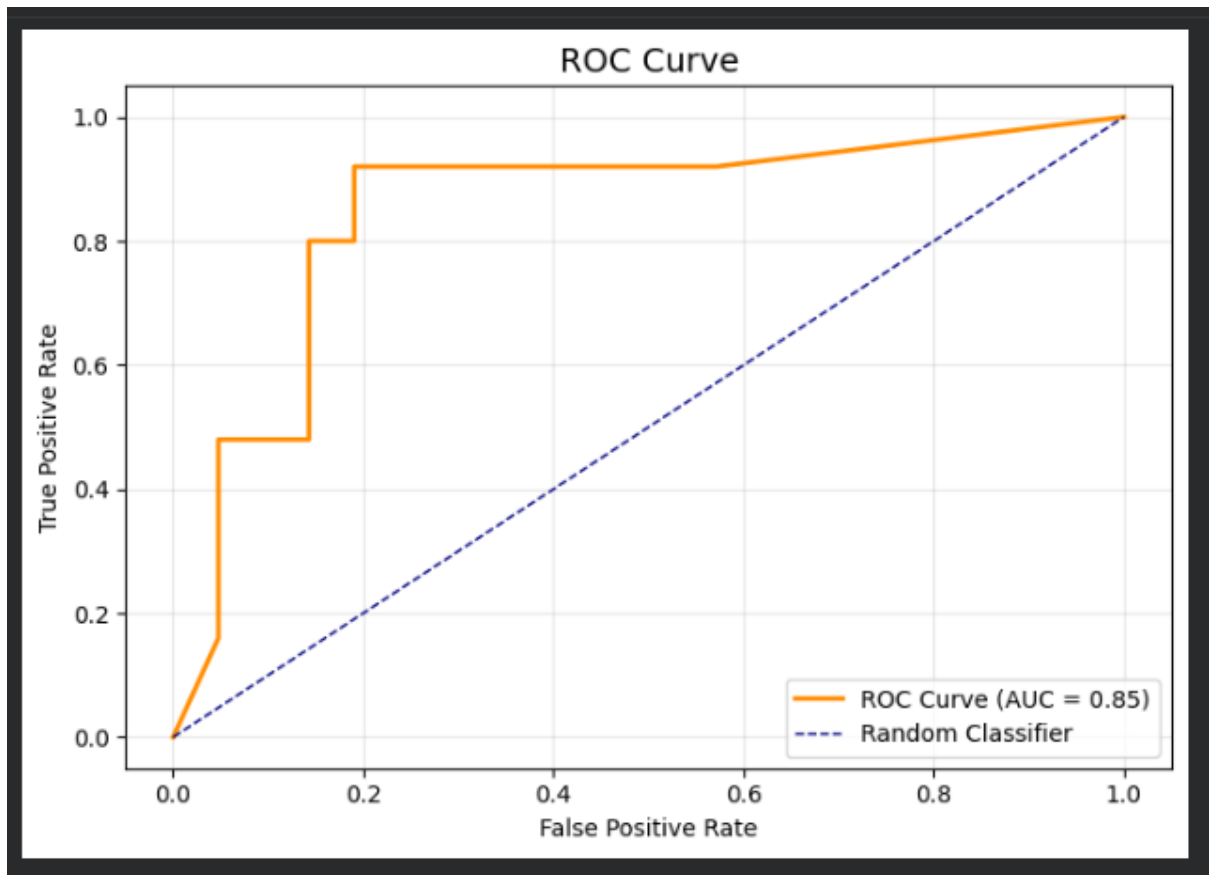


Figure 8: ROC Curve.

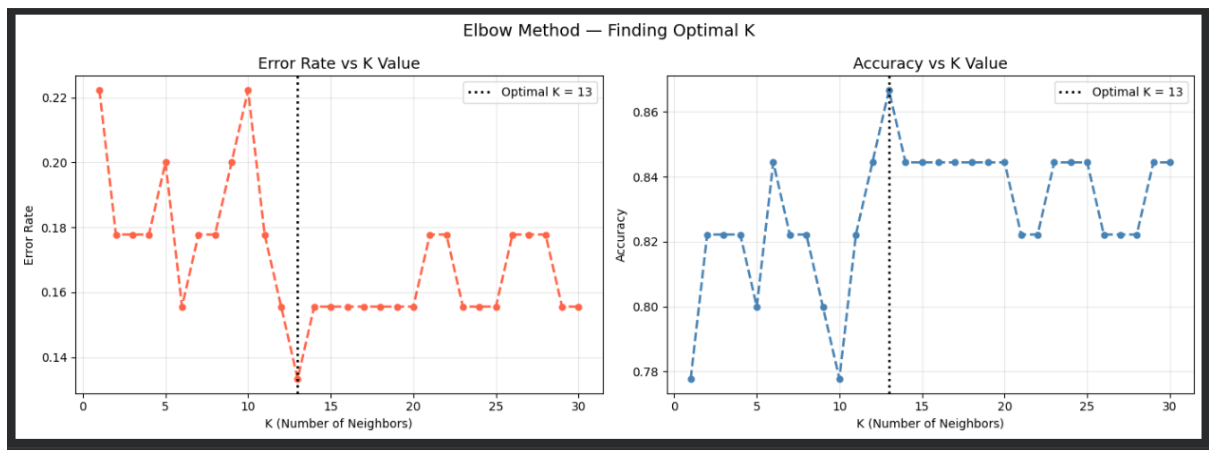


Figure 9: Elbow method plot.

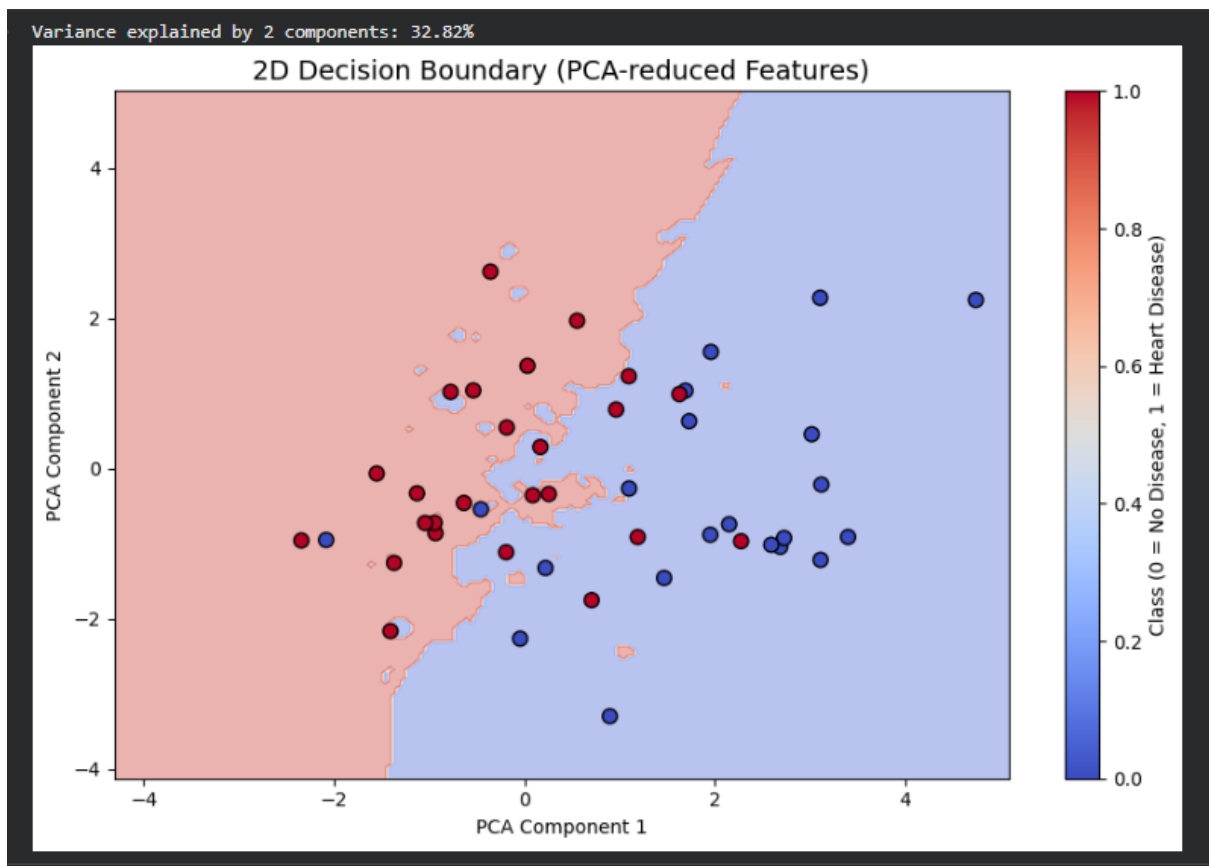


Figure 10: 2d decision boundary plot.